Salary Prediction Using Regression

Machine learning model to predict salaries based on professional and demographic features

Project Overview

Goal

Predict salary using ML based on experience, education, location, and demographics

Dataset

6,684 records from Kaggle with age, gender, education, job title, experience, country, race, senior level

Output

Streamlit web interface for instant predictions with automated preprocessing

Dataset Features

Numeric Features

- Age
- Years of Experience
- Senior Level

Categorical Features

- Gender
- Education Level
- Job Title
- Country
- Race

Project Architecture

Dataset
6,684 cleaned records

Data Preprocessing
Cleaning, encoding, scaling

Train-Test Split
80% training, 20% testing

Random Forest Model
Ensemble learning algorithm

Model Evaluation
MAE and R^2 metrics

Save Model
Joblib serialization

Streamlit Interface
User input and prediction

Data Preprocessing

01

Load Dataset

Pandas DataFrame from CSV file

02

Clean Target

Convert salary to numeric, handle invalid values

03

Remove Missing

Drop rows with NaN values

04

Feature Engineering

StandardScaler for numeric, OneHotEncoder for categorical

05

Column Transformer

Apply different preprocessing per column type

Machine Learning Model

Why Random Forest?

- Ensemble of decision trees
- Handles nonlinear relationships
- Reduces overfitting
- Works with mixed data types

Model Pipeline

Combined preprocessing and training ensures consistent predictions and prevents data leakage

Model Evaluation Results

97.2%

R^2 Score

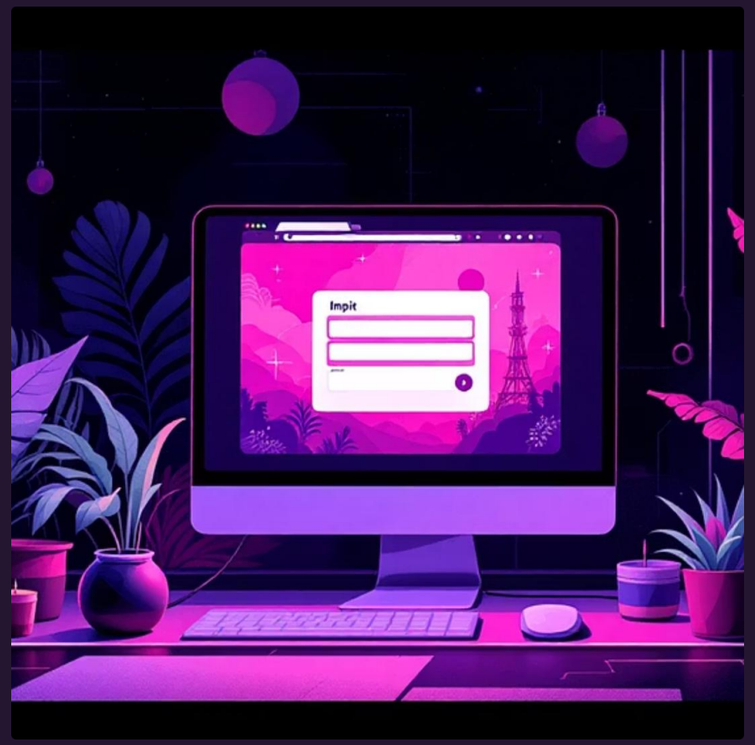
Model explains 97.2% of salary variance

3,844

MAE (Mean Absolute Error)

Average prediction error in salary units

Web Application



Streamlit Interface

User-friendly web app for input

Input Features

Age, experience, education, job, country

Instant Prediction

Model processes and displays result

Example Prediction

Input Provided

- Age: 30
- Experience: 6 years
- Job Title: Manager
- Country: USA

Prediction Result

₹101,000

Based on patterns from 6,684 training records

Technologies & Conclusion

Python & Libraries

Pandas, NumPy, Scikit-learn, Joblib

Streamlit

Web application interface

High Accuracy

R^2: 0.97, MAE: 3,844